

COMMUNITY ENGAGEMENT PROJECT (CEP)

SKILL BRIDGE

Community Employment and Skill Matching Portal

Submitted in Partial Fulfilment

of the Requirements for the Award of the Degree

of

BACHELOR OF SCIENCE

[To be added: Information Technology / Data Science]

SEMESTER V

By

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Roll Number: 24302D0074

Under the esteemed guidance of

[To be added: Guide name]

[To be added: Guide designation]

Department of Computing

VSIT | Vidyalankar School of
Information Technology

RE-ACCREDITED : GRADE 'A' BY NAAC

DEPARTMENT OF COMPUTING

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

(Autonomous College affiliated to University of Mumbai)

MUMBAI, 400037

MAHARASHTRA

2026 - 2027

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

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CERTIFICATE

I hereby certify that Varun Patil (Roll Number: 24302D0074), student of Vidyalankar School of Information Technology, studying in T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, has completed a C.E. project titled “Skill Bridge — Community Employment and Skill Matching Portal” at Khalsa College, Matunga, during the academic year 2026–27.

Internal Guide

Head of Department

Department of Computing

Internal Examiner

External Examiner

Date:



Principal

ORGANIZATION VISIT OBSERVATION SHEET

Attach the Organization Visit Observation Sheet duly signed by the reporting officer and mentor.

Organization: Khalsa College, Matunga

Student: Varun Patil | Roll Number: 24302D0074

[To be added: Signed observation sheet covering all three visits]

PROJECT GUIDE MEETING RECORD

Attach the Project Guide Meeting Record duly signed by the mentor.

Project: Skill Bridge — Community Employment and Skill Matching Portal

Student: Varun Patil | Roll Number: 24302D0074

[To be added: Signed guide meeting record]

DECLARATION

I, Varun Patil, of T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, Vidyalkar School of Information Technology, present this report on “Skill Bridge — Community Employment and Skill Matching Portal” as the record of our CEP work for 2026–27.

The fieldwork consisted of three visits to Khalsa College, Matunga. Our team collected real student responses by manually circulating a Google Form. The technical descriptions in this report concern the shared application developed for the project.

Sources used in preparing the report are identified in the references. Supporting forms and signatures are to be attached in their prescribed locations.

Signature of the Student with Date: _____

Varun Patil

Roll Number: 24302D0074

ACKNOWLEDGEMENT

I thank Vidyalankar School of Information Technology and the University of Mumbai for the opportunity to complete a project with a community setting. The work gave practical meaning to the development and database concepts studied in our course.

I acknowledge our faculty mentor, [To be added: Guide name], and thank Khalsa College, Matunga, for hosting the project visits. I am also grateful to the students who shared information through the Google Form.

My teammates, Dhiraj Tendulkar, Vidhi Mahato and Gaurav Mhatre, contributed to the common project effort. I thank them and the faculty members who supported our work.

ABSTRACT

Our Community Engagement Project resulted in Skill Bridge, a portal for student employment preparation and placement coordination. The team visited Khalsa College, Matunga, three times and gathered real student responses by manually distributing a Google Form. The project's central concern is continuity: students should be able to move from finding an opportunity to taking action and receiving a response, while staff should have an organised view of the work requiring attention.

The application separates student, placement coordinator and administrator access. Its functions include profile management, opportunity discovery, explained matching, application tracking, workshops, feedback, guidance and research summaries. Coordinator actions update shared operational records, while administrative tools support configuration and data recovery. Individual survey responses are restricted, and public reporting uses aggregate information.

React, TypeScript, Vite, Tailwind CSS and Recharts form the frontend stack. The backend uses Express and SQLite. Matching is implemented as a deterministic scoring function that compares selected profile and listing attributes. Because it is a relevance aid rather than a predictive model, the interface also provides reasons and eligibility information. Authentication, server-side permissions, validation and relational constraints support the main workflows.

The latest recorded test run passed 157 checks across four suites, and both the production build and TypeScript verification succeeded. Fresh production serving, administrator login, an authorized update and logout were included in the checks. These results describe technical behaviour under the test conditions. They do not demonstrate live institutional adoption, performance under load or a change in employment outcomes. The project offers a usable foundation for coordinated placement support. Original survey totals, photographs, signed feedback and user-acceptance results are to be attached as the remaining fieldwork evidence.

TABLE OF CONTENT

CHAPTER 1: INTRODUCTION	1
1.1 Project Background.....	1
1.2 Problem Statement.....	1
1.3 Need for the Project	1
1.4 Objectives	2
1.5 Scope.....	2
1.6 SDG Alignment	2
CHAPTER 2: REQUIREMENT ANALYSIS AND LITERATURE REVIEW	3
2.1 Methodology Adopted	3
2.2 Stakeholder Interactions.....	3
2.3 Domain Overview	4
2.4 Existing Process Analysis	4
2.5 Existing Solutions and Related Work	4
2.6 Technologies Studied.....	4
2.7 Research Gap and Need for Proposed Solution	5
2.8 Challenges Identified	5
CHAPTER 3: PROPOSED SOLUTION	6
3.1 Solution Overview	6
3.2 System Architecture.....	6
3.3 Process Flow Diagram	6
3.4 Module Description	6
3.5 Advantages of Proposed Solution.....	7
CHAPTER 4: SYSTEM DESIGN	8
4.1 System Design Overview.....	8
4.2 Use Case Diagram.....	8
4.3 Workflow Diagram	9
4.4 Module Design.....	9
CHAPTER 5: PROJECT IMPLEMENTATION	11
5.1 Development Methodology	11
5.2 Module-wise Implementation	11
5.3 User Interface Solution	12
5.4 Screenshots of Developed Solution	13
CHAPTER 6: TESTING AND VALIDATION	14
6.1 Test Cases	14

6.2 Test Results	15
6.3 User Acceptance Testing	15
6.4 Feedback from Stakeholders	15
6.5 Performance Evaluation	15
CHAPTER 7: RESULTS AND IMPACT ASSESSMENT	16
7.1 Deliverables Submitted	16
7.2 Results Achieved	16
7.3 Impact on Organization	16
7.4 SDG Contribution	16
CHAPTER 8: CONCLUSION AND FUTURE SCOPE	17
8.1 Conclusion	17
8.2 Key Findings	17
8.3 Recommendations	17
8.4 Future Enhancements	17
REFERENCES	18
Research Papers	18
Books Referred	18
Website	18
ANNEXURES	19
Annexure A: Field Visit Photographs	19
ANNEXURE B: SURVEY QUESTIONNAIRE	22
ANNEXURE C: INTERVIEW QUESTIONS	24
ANNEXURE D: REPORTING OFFICER FEEDBACK FORM	25
ANNEXURE G: SOURCE CODE REPOSITORY	26
ANNEXURE H: USER MANUAL	27

TABLE OF FIGURES

Figure 1: System Architecture of Skill Bridge (placeholder)	6
Figure 2: Process Flow of Skill Bridge (placeholder).....	6
Figure 3: Use Case Diagram of Skill Bridge (placeholder)	8
Figure 4: Application and Workshop Workflow of Skill Bridge (placeholder)	9
Figure 5: Key Components of the Skill Bridge Database (placeholder).....	10
Figure 6: Student Workspace of Skill Bridge (placeholder).....	13
Figure 7: Placement Coordinator Workspace of Skill Bridge (placeholder)	13
Figure 8: Research Summary View of Skill Bridge (placeholder)	13
Figure 9: Visit 1 at Khalsa College, Matunga.....	19
Figure 10: Visit 2 at Khalsa College, Matunga.....	20
Figure 11: Visit 3 at Khalsa College, Matunga.....	21

TABLE OF TABLES

Table 1: Project team	1
Table 2: Fieldwork record.....	3
Table 3: Technologies used	4
Table 4: Functional modules.....	7
Table 5: Actors and access.....	8
Table 6: Database entities	9
Table 7: Matching score components	10
Table 8: Implementation mapping.....	11
Table 9: Representative test cases.....	14
Table 10: Recorded automated results	15
Table 11: Repository contents.....	26

CHAPTER 1: INTRODUCTION

1.1 Project Background

Skill Bridge is the web application developed for our Community Engagement Project. All three visits took place at Khalsa College, Matunga, and the team collected genuine student responses by distributing a Google Form manually. The portal addresses employment preparation and placement coordination through connected student and staff workspaces. My report explains how the project was organised, how its features operate and what the available technical evidence establishes.

Table 1: Project team

Member	Roll number
Dhiraj Tendulkar	24302A0004
Vidhi Mahato	24302A0028
Gaurav Mhatre	24302A0067
Varun Patil	24302D0074

1.2 Problem Statement

Our project focuses on the practical gap between seeing an opportunity and receiving useful placement support. Students need relevant listings and visible progress; coordinators need structured records for decisions, training and guidance. The application was designed to support that whole sequence. Claims about the frequency or scale of these difficulties require the field-response analysis, which is [To be added].

1.3 Need for the Project

The value of the project lies in making placement support understandable and traceable. A student should know what is available, what has been submitted and what response has been received. Staff should have an organised queue of work rather than relying on the student interface. The portal supports these needs without presenting a matching score as a guarantee of selection.

1.4 Objectives

Our objectives were framed around the tasks the portal should support. The distinction below keeps the student-facing purpose clear while recognising the need for reliable administration and validation.

Primary Objectives

- Provide a distinct student workspace and coordinator workspace with server-enforced access.
- Support discovery, saving and application for opportunities with visible status updates.
- Connect workshop participation, feedback and guidance with coordinator follow-up.

Secondary Objectives

- Explain opportunity matching through understandable rules and reasons.
- Preserve related records and reject invalid or unauthorized changes.
- Provide aggregate research reporting while protecting individual responses.
- Maintain repeatable tests and practical operating instructions.

1.5 Scope

Skill Bridge covers the technical portal, its database, the documented workflows and the fieldwork setting. It supports placement preparation and coordination rather than replacing an employer's assessment process. Matching remains advisory, and no verified claim is made about placement-rate improvement. The final hosting location, approved operational owner and formal handover details are [To be added].

1.6 SDG Alignment

Two goals provide a suitable framework for the project: SDG 4, Quality Education, and SDG 8, Decent Work and Economic Growth. Workshops, guidance and explained matching connect learning needs with career preparation. The application can support progress toward these goals, but a verified community-impact measurement remains a separate evaluation task.

CHAPTER 2: REQUIREMENT ANALYSIS AND LITERATURE REVIEW

2.1 Methodology Adopted

Our work had two connected parts. Fieldwork at Khalsa College, Matunga, provided the setting for manually distributing the Google Form and collecting student inputs. The development work converted placement-support requirements into software features and checked them repeatedly. Dates, attendance and verified survey totals are [To be added]. This separation helps avoid treating application test records as field-survey results.

Table 2: Fieldwork record

Item	Confirmed information or record
Organization	Khalsa College, Matunga
Visits	Three visits, all at the same college
Collection method	Google Form manually circulated among students
Visit dates	[To be added]
Confirmed response count	[To be added]
Google Form link and response export	[To be added]
Verified findings and response summary	[To be added]

2.2 Stakeholder Interactions

The stakeholder view starts with the students who received our Google Form. The portal additionally provides staff workspaces for coordination and administration. All four team members contributed to the shared CEP project; individual task-allocation records are [To be added]. Staff names and attributable interview statements will require their original records before inclusion.

2.3 Domain Overview

The project domain is the coordination of career opportunities and preparation within a student community. Profiles describe the applicant, listings describe available roles, and applications record participation. Training, feedback and guidance complete the support cycle. The portal provides an organised workflow within this domain without claiming to make recruitment decisions.

2.4 Existing Process Analysis

The existing process that can be stated with certainty is manual circulation of the Google Form among students. The project extends beyond collection by providing accounts, application tracking and staff responses. It would be inaccurate to attribute undocumented procedures or failures to the college; its approved baseline workflow is [To be added].

2.5 Existing Solutions and Related Work

Google Forms was part of our requirement collection, but the developed website has a broader operational purpose. It provides continuing access to profiles, listings, decisions and support requests. No comparative performance claim against another product is made here. Details of any research papers, books or formal product evaluations are [To be added].

2.6 Technologies Studied

The implementation combines React components with an Express API and a SQLite database. TypeScript checks the contracts between data and code, while Vite prepares the frontend build. Tailwind CSS and Recharts address interface styling and research visualisation. The stack is described according to its use in Skill Bridge rather than through generic technology claims.

Table 3: Technologies used

Technology	Role in Skill Bridge
React	Component-based frontend and role-specific pages
TypeScript	Typed models and compile-time checks for frontend and server
Vite	Frontend development server and production bundling
Tailwind CSS	Interface layout and responsive styling

Technology	Role in Skill Bridge
Recharts	Research charts and visual summaries
Node.js and Express	Runtime, API routes, authentication and authorization
SQLite through node:sqlite	Persistent relational records, foreign keys and transactions
tsx and oxlint	TypeScript execution and lint checks
Google Forms	Manual circulation and collection of real student field responses

2.7 Research Gap and Need for Proposed Solution

Our solution addresses the transition from understanding student needs to supporting concrete next steps. The portal links profiles and listings with applications, training and guidance. This is the project's implementation gap, not a claim that no comparable systems exist. Any broader research comparison requires verified literature.

2.8 Challenges Identified

Reliable coordination required more than adding pages. We had to account for access control, invalid inputs, duplicate actions and safe database startup. A further requirement was to protect individual survey responses while keeping public summaries usable. Specific issues experienced during survey collection are [To be added] from the original records.

CHAPTER 3: PROPOSED SOLUTION

3.1 Solution Overview

The portal organises placement support around the people using it. Student actions create records, coordinator actions update their progress, and administrative tools protect the system's configuration and data. The solution therefore combines an understandable interface with server-enforced control over what each account can do.

3.2 System Architecture

The frontend communicates with the server through HTTP API calls. Session and role checks determine whether a request can proceed, then the server performs the necessary database operation. The interface updates from the returned result. This arrangement centralises authority and persistence instead of relying on browser-only state for important records.

[To be added]

Figure 1: System Architecture of Skill Bridge (placeholder)

3.3 Process Flow Diagram

A typical transaction begins when a signed-in user selects an action. The API checks the session, role and input before changing data. The result then returns to the interface. For applications, this sequence continues with coordinator review and a student-visible decision, forming a traceable cycle.

[To be added]

Figure 2: Process Flow of Skill Bridge (placeholder)

3.4 Module Description

The solution is divided into modules with specific responsibilities. Shared services handle communication and state, while the functional modules manage their own records. The table below connects each module with the activity it supports.

Table 4: Functional modules

Module	Responsibility
Authentication and access	Sessions, login/logout, role checks and account isolation
Student profile and matching	Profile preferences, skills and explained opportunity relevance
Opportunities and applications	Listing discovery, saved openings, submission and review status
Workshops	Session management, capacity, enrolment and attendance
Feedback and guidance	Student requests and coordinator responses
Research	Restricted response records and public aggregate reporting
Study and visit records	Three-visit documentation, action plans and evidence fields
Administration	Protected content, approved diagnostics and backup/restore

3.5 Advantages of Proposed Solution

Skill Bridge's advantages come from its connected workflow, understandable matching and differentiated permissions. The interface can show a student's own progress while allowing coordinators to manage the broader queue. These features provide a basis for better coordination; their institutional impact must be measured through subsequent use.

CHAPTER 4: SYSTEM DESIGN

4.1 System Design Overview

The internal design combines typed application models with relational storage. It avoids treating a visible button as an authorization mechanism: the server independently checks each protected request. Validation, uniqueness constraints and transactions are used where invalid or repeated actions could create inconsistent records.

4.2 Use Case Diagram

The principal actors have different responsibilities. A visitor sees public information; a student applies or requests support; a coordinator reviews operational work; an administrator maintains the system. The permissions table defines these boundaries independently of the presentation layer.

Table 5: Actors and access

Actor	Permitted scope
Public visitor	Permitted public pages, opportunity information and aggregate research
Student	Own profile, saved listings, applications, enrolments, feedback and guidance
Placement Coordinator	Operational management, decisions, rosters and restricted research
Administrator	Protected configuration, diagnostics, backups and authorized management

[To be added]

Figure 3: Use Case Diagram of Skill Bridge (placeholder)

4.3 Workflow Diagram

The workflow model places checks before persistence. Authentication establishes identity; authorization establishes permission; validation checks the requested change. Application review and workshop participation then operate on stored records. The interface displays the server result rather than assuming every submitted action succeeded.

[To be added]

Figure 4: Application and Workshop Workflow of Skill Bridge (placeholder)

4.4 Module Design

The database model and matching process are important parts of the module design. Relational tables retain user actions, while a rule-based function produces an understandable opportunity score. The function helps exploration but leaves detailed eligibility and selection decisions to the stated recruitment conditions.

Database records and relationships

Table 6: Database entities

Entity or group	Purpose
users; sessions	Account identities, roles and active authentication sessions
student_profiles	Student-specific skills, interests and academic information
opportunities; applications	Listings linked to student submissions and review decisions
saved_opportunities	Student-specific saved listings
training_sessions; workshop_enrollments	Workshop information and participation relationships
feedback_submissions; guidance_requests	Requests and staff follow-up
survey_responses	Structured responses with provenance and controlled access

Entity or group	Purpose
employer_outreach	Staff-managed employer contact records
system_documents; app_metadata	Content documents and database initialisation state

[To be added]

Figure 5: Key Components of the Skill Bridge Database (placeholder)

Opportunity matching process

Table 7: Matching score components

Component	Points	Implemented rule
Department	40	Awarded when the listing includes the profile department
Preferred role	25	Case-insensitive containment between a preferred role and listing title
Skill overlap	Up to 20	Seven points per overlapping listed skill, capped at 20
Location	10	Preferred location matches, or the listing uses remote work
Academic cutoff	5	Entered percentage meets a detected two-digit percentage cutoff

The score is the sum of the awarded components, bounded between 0 and 100. Skill points are calculated as $\min(20, 7 \times \text{number of matching listing skills})$. A score of 75 or more is labelled High Fit; 50–74 is Moderate Fit; 30–49 is Partial Fit; lower scores are General Opportunity.

Text comparisons use case-insensitive containment, so the result depends on the terms entered in the profile and listing. The eligibility check detects a two-digit percentage cutoff and returns an eligibility caution where appropriate. It is not a complete eligibility parser. Students must review the original criteria even when the fit score is high.

CHAPTER 5: PROJECT IMPLEMENTATION

5.1 Development Methodology

Implementation progressed through a sequence of feature work, inspection and regression checks. The aim was to preserve working behaviour while correcting specific problems. Production compilation and isolated tests provided evidence after changes, and the real database was protected from test resets.

5.2 Module-wise Implementation

The implementation links user-facing screens to protected operations. React renders the workspace, the API service handles requests, and Express checks identity, role and input. The database stores accepted updates for later retrieval. The same pattern is used across applications, workshops, feedback and guidance, with workflow-specific checks.

Table 8: Implementation mapping

Source area	Implementation role
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Authentication and data integrity

Passwords are hashed with bcrypt and a salt. Session tokens are carried in an HttpOnly cookie, and protected routes verify the session and role. Public registration is restricted to the student role. Database uniqueness constraints help prevent duplicate applications and enrolments. Restore operations validate supported content and relationships before transactional updates.

Research summaries

The reporting process groups responses by the requested filters and computes counts, distributions and rating summaries. Non-rated responses such as “Have not attended” are not treated as numeric satisfaction scores. Public reporting uses aggregate endpoints; individual responses are limited to authorized staff. The actual field-response totals and numerical findings are [To be added].

5.3 User Interface Solution

The frontend translates the access model into clear navigation. Student actions, coordinator decisions and administrator controls appear in their respective workspaces. Loading and error handling support the interaction around API requests. A recorded browser walkthrough across desktop and mobile devices is [To be added].

5.4 Screenshots of Developed Solution

The screenshot section is reserved for evidence from the actual portal. Insert the listed views after checking that they match the submitted version and do not expose confidential account or survey information.

(add picture)

Figure 6: Student Workspace of Skill Bridge (placeholder)

(add picture)

Figure 7: Placement Coordinator Workspace of Skill Bridge (placeholder)

(add picture)

Figure 8: Research Summary View of Skill Bridge (placeholder)

CHAPTER 6: TESTING AND VALIDATION

6.1 Test Cases

The project was checked through acceptance, workflow, lifecycle and serving suites. Together they cover calculations, protected operations, database initialisation and the production server. These automated checks use controlled data; they are distinct from acceptance by students or college staff.

Table 9: Representative test cases

Area	Scenario	Expected result
Authentication	Invalid credentials and unauthenticated requests	Rejected
Role authorization	Student requests staff-only resources	Rejected
Account isolation	One student requests another student's records	Private records not exposed
Application lifecycle	Submit, reject duplicate, record coordinator decision	Expected records and status retained
Workshop lifecycle	Enrol, reject duplicate, cancel, enforce capacity	Participation rules enforced
Research privacy	Public/student access to raw responses	Blocked; aggregate reporting available
Content and restoration	Malformed documents and invalid relationships	Rejected; valid restoration supported
Diagnostics	Client-supplied SQL and approved query identifiers	Raw SQL blocked; approved queries work
Production lifecycle	Fresh database, restart and admin provisioning	Startup and provisioning checks passed

6.2 Test Results

The recorded verification completed successfully across acceptance, workflow/security, production lifecycle and serving checks. TypeScript and the production build also passed. Lint completed with warnings. No numerical claim about real-user satisfaction or employment outcomes follows from these software checks.

Table 10: Recorded automated results

Suite	Checks	Result
Acceptance	37	Passed
Workflow and security	97	Passed
Production lifecycle	14	Passed
Production serving	9	Passed
Total	157	Passed

6.3 User Acceptance Testing

The remaining UAT evidence should identify the users, tasks, results and approval status. These details are [To be added]. Students and staff should verify the workflows in the browser, including error messages and recovery, before a formal acceptance claim is made.

6.4 Feedback from Stakeholders

The final feedback section should contain the findings drawn from the collected responses and any signed staff assessment. Those details are [To be added]. The blank form in Annexure D provides space for that assessment while leaving the authority to approve the work with the reporting officer.

6.5 Performance Evaluation

The present evidence supports functional validation. It does not provide an accuracy percentage for matching or a measured server-capacity figure. Timing, memory and load-test results are [To be added]. Any later performance claim should identify the environment, workload and measurement method.

CHAPTER 7: RESULTS AND IMPACT ASSESSMENT

7.1 Deliverables Submitted

The completed technical outputs are the portal implementation, persistent data model, checks and operating instructions. Supporting documentation includes this report and its annexures. Organisation-level handover, repository publication and hosting details are [To be added] where they have not been supplied.

7.2 Results Achieved

The main outcome is an implemented system that supports the intended student–coordinator cycle. Local production checks and the full test command passed in the latest review. This supports readiness for further browser and host validation. It does not establish a measured increase in placements or institutional adoption.

7.3 Impact on Organization

Skill Bridge can assist the college by keeping opportunities, participation and support requests in an organised workflow. Whether this improves everyday practice must be assessed after use. Adoption records and before-and-after measures are [To be added]; the available technical results alone cannot establish that impact.

7.4 SDG Contribution

The SDG connection is visible in the project’s purpose and functions. Skill-development opportunities support SDG 4, while career information and application support relate to SDG 8. The report records this contribution at the implementation level and leaves measured long-term impact for subsequent evaluation.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Skill Bridge provides a concrete outcome for the CEP work: a portal with distinct access, persistent records and verified local production behaviour. The project is grounded in three visits and real student data collection at Khalsa College. Its continuing value will depend on careful deployment, maintained information and documented feedback from users.

8.2 Key Findings

The strongest technical lesson is the need to verify complete workflows rather than individual pages alone. Data integrity, account isolation and startup configuration affect every user journey. The project also requires a clear distinction between real survey evidence and controlled software tests when presenting results.

8.3 Recommendations

Operational preparation should include a responsible administrator, a maintained backup schedule and a tested recovery process. The team should complete browser verification and obtain formal feedback. Survey totals, dates and signatures should be filled from verified records, and confidential response data should remain protected.

8.4 Future Enhancements

The project can be extended through user-approved notifications, adaptable matching criteria and longer-term reporting. These additions should follow evidence of need. Future work should also measure accessibility and performance under realistic use, while retaining the existing boundaries around personal and survey data.

REFERENCES

Research Papers

[To be added: Verified research papers actually consulted, formatted in MLA style.]

Books Referred

[To be added: Books actually consulted, with verified MLA bibliographic details.]

Website

[To be added: Websites actually consulted and the topics referred to, with verified URLs and access details.]

Project source material:

Skill Bridge Project Team. Skill Bridge: Community Employment and Skill Matching Portal. Project source code, 2026.

Skill Bridge Project Team. “Project Setup and Verification.” README.md, 2026.

Skill Bridge Project Team. “Opportunity Matching.” src/utis/matching.ts, 2026.

Skill Bridge Project Team. “Survey Questionnaire.” src/data/defaultSurveySeed.ts, 2026.

Skill Bridge Project Team. “Verification Suites.” scripts/verify-acceptance.ts, scripts/test-workflows.ts, scripts/test-production-startup.ts, and scripts/verify-production-serving.ts, 2026.

ANNEXURES

The annexures retain the labels prescribed in the supplied format: A, B, C, D, G and H.

Annexure A: Field Visit Photographs

Visit 1 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 9: Visit 1 at Khalsa College, Matunga

Visit 2 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 10: Visit 2 at Khalsa College, Matunga

Visit 3 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 11: Visit 3 at Khalsa College, Matunga

ANNEXURE B: SURVEY QUESTIONNAIRE

Real student responses were collected through the Google Form distributed by the team. This annexure presents the project's stored questionnaire. The circulated form link and final version confirmation are [To be added].

1. Which department or course are you studying in?

Response options: Computer Science (CS); Information Technology (IT); Data Science (DS).

2. What is your current year of study?

Response options: First; Second; Third; Fourth; Other.

3. What job role or career domain are you mainly interested in?

Response options: Open text response.

4. How many campus recruitment drives offered roles relevant to your preferred domain during the current academic year?

Response options: 0; 1–2; 3 or more; Not sure.

5. Is the placement cell currently meeting your placement-related needs?

Response options: Yes; Partly; No; Not enough experience to judge.

6. How often do you receive placement information early enough to prepare and apply?

Response options: Always; Sometimes; Rarely; Never; Not applicable.

7. How useful has the placement training you attended been for your preferred role?

Response options: 1–5, where 1 is Not useful and 5 is Very useful; Have not attended.

8. What is the biggest barrier you face in getting a suitable placement?

Response options: Limited relevant roles; Eligibility restrictions; Skill gaps; Late information; Interview preparation; Location or pay mismatch; Other; No major barrier.

9. Which placement support do you need most urgently?

Response options: Relevant employer connections; Technical training; Aptitude preparation; Resume and interview help; Career guidance; Timely alerts; Internship support; Other.

10. How satisfied are you with overall placement support?

Response options: 1–5, where 1 is Very dissatisfied and 5 is Very satisfied; Not enough experience to judge.

11. Briefly describe your experience with the placement process.

Response options: Open text response.

12. What one change would most improve placement support for you?

Response options: Open text response.

ANNEXURE C: INTERVIEW QUESTIONS

The project includes an interview guide for placement-coordination discussions. The questions below reproduce that guide. Interview dates, respondent identity, consent and recorded answers are [To be added]. Illustrative answers in the software are not reproduced as actual staff statements.

1. How are student preferences and skills collected?
2. How many relevant employers and roles were available by department?
3. What explains differences in opportunities?
4. How are new employers approached?
5. What makes specialist employers difficult to attract?
6. How are notices received and published?
7. How are training needs identified?
8. How is training effectiveness measured?
9. What alternatives exist for students excluded by eligibility criteria?
10. Are duties, location, and pay details checked before publication?
11. What staffing, time, or budget constraints exist?
12. Which concerns are already being addressed?
13. What additional actions are feasible, with owners and timelines?
14. What can Skill Bridge support?
15. How is student feedback reviewed and answered?

ANNEXURE D: REPORTING OFFICER FEEDBACK FORM

Blank form for assessment of the prepared project

Project: Skill Bridge — Community Employment and Skill Matching Portal

Organization: Khalsa College, Matunga

Student: Varun Patil | Roll Number: 24302D0074

Reporting officer name: _____

Designation: _____

Date of review: _____

Relevance of the project to student/community needs

Usefulness and clarity of the developed solution

Observations on the demonstration

Suggested improvements

Overall assessment and acceptance status

Reporting officer signature: _____

Organization stamp, if required: _____

ANNEXURE G: SOURCE CODE REPOSITORY

Repository link: [To be added]

Submitted version or commit identifier: [To be added]

Contents uploaded and verified by: [To be added]

The following inventory identifies the project contents to include in the submission. It does not confirm that a repository has already been published.

Table 11: Repository contents

Source area	Contents
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Exclude passwords, private environment files, real student records and local database backups from any public repository. Provide authorised access to sensitive evidence separately.

ANNEXURE H: USER MANUAL

This guide summarises routine operation of Skill Bridge. The available controls depend on the account role. The final portal URL and operational contact are [To be added].

Student workflow

1. Authenticate as a student and confirm that profile information is current.
2. Explore relevant opportunities and open their full details.
3. Compare the explanation with the original eligibility conditions.
4. Save or apply to the opportunity and monitor the stored application status.
5. Manage workshop participation through the available enrolment controls.
6. Use the support forms for feedback or guidance and end the session by logging out.

Coordinator workflow

1. Log in to the placement coordinator account.
2. Manage the published opportunities and workshop details.
3. Inspect submitted applications and store decisions with any relevant feedback.
4. Use the enrolment roster for participation and attendance work.
5. Follow up on guidance and feedback queues.
6. Verify successful updates and close the session through logout.

Administrator workflow

1. Access the administrative workspace using the assigned credentials.
2. Maintain system content and use the approved diagnostics.
3. Back up the database before sensitive maintenance.
4. Submit a suitable backup for restoration and inspect the returned outcome.
5. Restrict access to backup files and administrator credentials.

Local setup and production preparation

Install a Node.js version compatible with the project dependencies and native SQLite support, then run npm install in the project folder.

Use `npm run dev` for local development. The configured API and frontend addresses should be read from the terminal output.

Run `npm run typecheck`, `npm run build` and `npm test` before preparing a release.

For production, set `NODE_ENV=production`, a persistent `DB_PATH` and `CLIENT_URL` matching the actual website origin. Configure HTTPS and the intended reverse proxy.

Provision an administrator with explicit credentials using `npm run admin:provision -- <username> <password>`. Use protected credential handling in the deployment environment.

Run `npm start` after the build. Verify login, authorised changes and direct route refreshes on the actual host.

Use `npm run backup:db` for the database backup utility. Test recovery with a disposable copy before relying on a backup. Do not run demo reset commands against the real database.

If an operation fails

When a request fails, inspect the current state first to avoid duplicate work. Check connectivity and permissions, and report continuing problems with the exact action and message. Do not include private records in a public error report.